

IDP - PHASE 2 HARDWARE PART

Phase 1 Suggestions

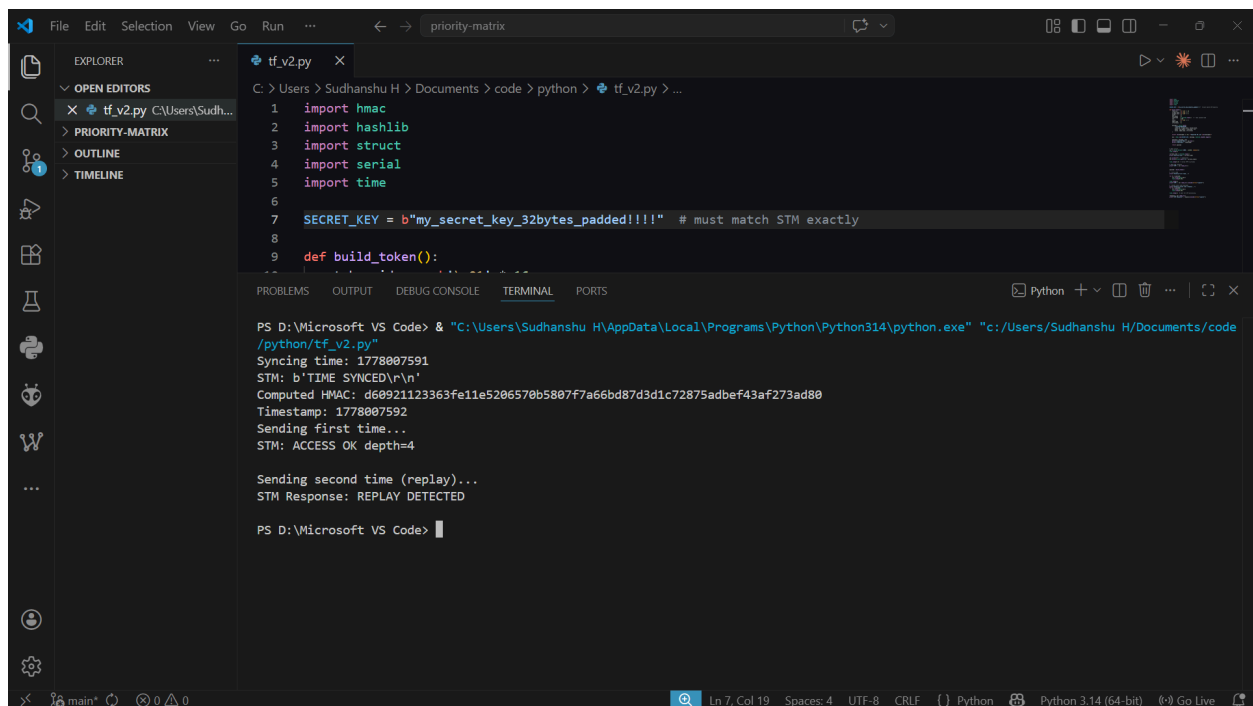
Tools and Techniques

Tools - STM32C071RB Nucleo Board, STM32 CubeIDE, STM32CubeMX, (+pyserial, hmac lib etc on software end for integration and encryption)

Work Progress

- Configured USART channel for communication with software via USB.
- Allocation of separate memory space for secure keys.
- Setup of HMAC Engine to run SHA256 on stored key, comparison of the received token with the obtained token as output of HMAC.
- Readout Protection Level 1 enabled, Level 2 used when product is deployed.

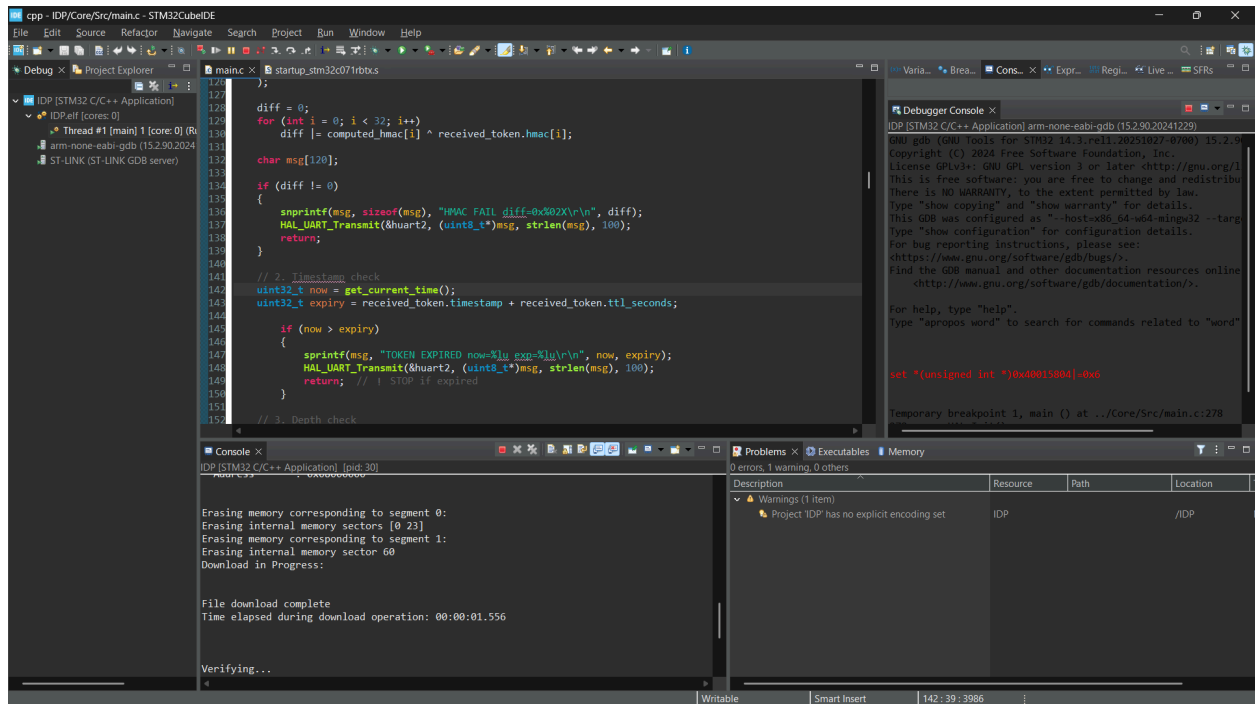
Partial Results



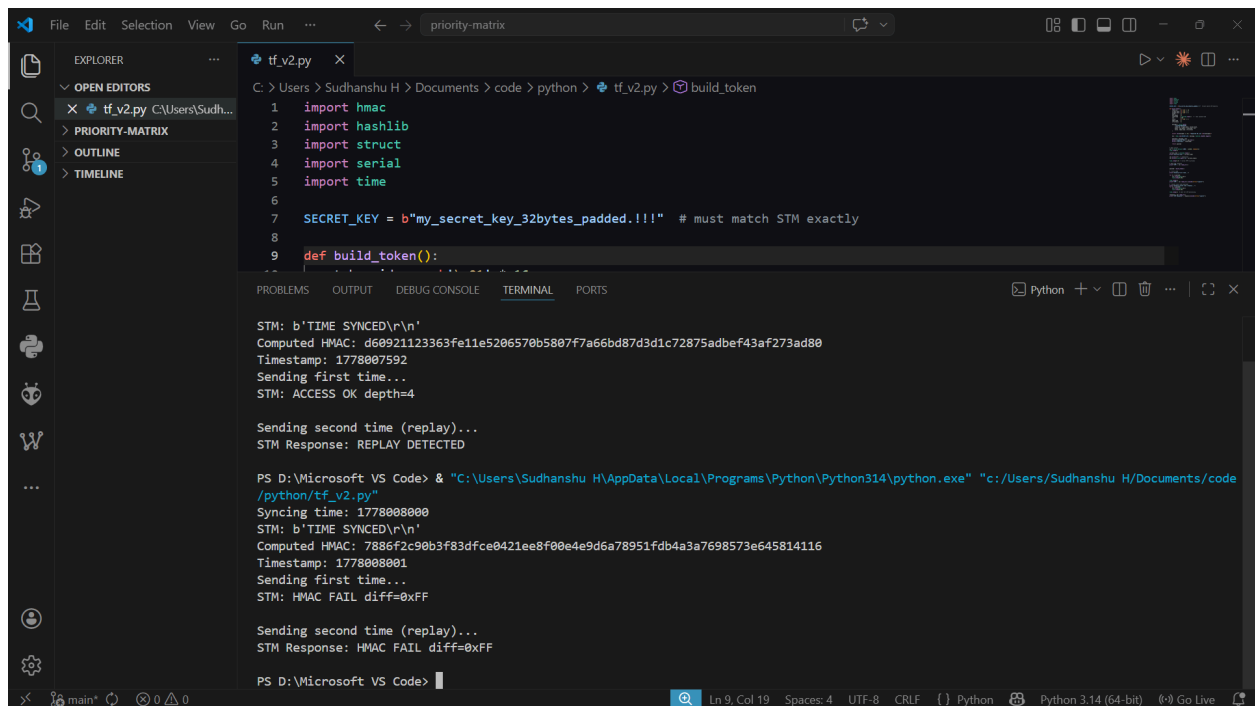
The screenshot shows a VS Code editor with a Python script named `tf_v2.py` open. The script imports `hmac`, `hashlib`, `struct`, `serial`, and `time`. It defines a `SECRET_KEY` and a `build_token()` function. The terminal output shows the script being executed, displaying the sync time, the generated token, the computed HMAC, and the timestamp. The first transmission is accepted as valid, but the second identical transmission is rejected due to nonce reuse.

```
1 import hmac
2 import hashlib
3 import struct
4 import serial
5 import time
6
7 SECRET_KEY = b"my_secret_key_32bytes_padded!!!!" # must match STM exactly
8
9 def build_token():
10     ...
11
12 PS D:\Microsoft VS Code> & "C:\Users\Sudhanshu H\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Sudhanshu H/Documents/code/python/tf_v2.py"
Syncing time: 1778007591
STM: b'TIME SYNCED\r\n'
Computed HMAC: d60921123363fe11e5206570b5807f7a66bd87d3d1c72875adbef43af273ad80
Timestamp: 1778007592
Sending first time...
STM: ACCESS OK depth=4
Sending second time (replay)...
STM Response: REPLAY DETECTED
PS D:\Microsoft VS Code>
```

The system successfully synchronizes time with the STM32, generates a valid HMAC, and sends a secure token. The first transmission is accepted as valid, confirming integrity and freshness. The second identical transmission is rejected due to nonce reuse, demonstrating effective replay attack protection and correct implementation of security checks.



The STM32CubeIDE debug session shows successful code compilation, flashing, and execution. The program verifies HMAC integrity, performs timestamp validation, and prepares to enforce expiration checks. Debugger and console outputs confirm proper memory programming and runtime flow, indicating the secure token validation logic is correctly deployed and ready for testing.



The system detects a mismatch between the received HMAC and the locally computed value, indicating the token has been altered or forged. As a result, the request is immediately rejected, ensuring data integrity and authenticity by preventing unauthorized or tampered tokens from being accepted by the hardware security layer.

Work to be done

- Ensuring byte-level consistency, accurate sampling rates, processing speed, synchronisation, preventing partial packet parsing, etc. when interfaced with software.
- Thorough testing to look for corner cases where the system fails on the hardware end.